D1.3 Mutation and gene editing

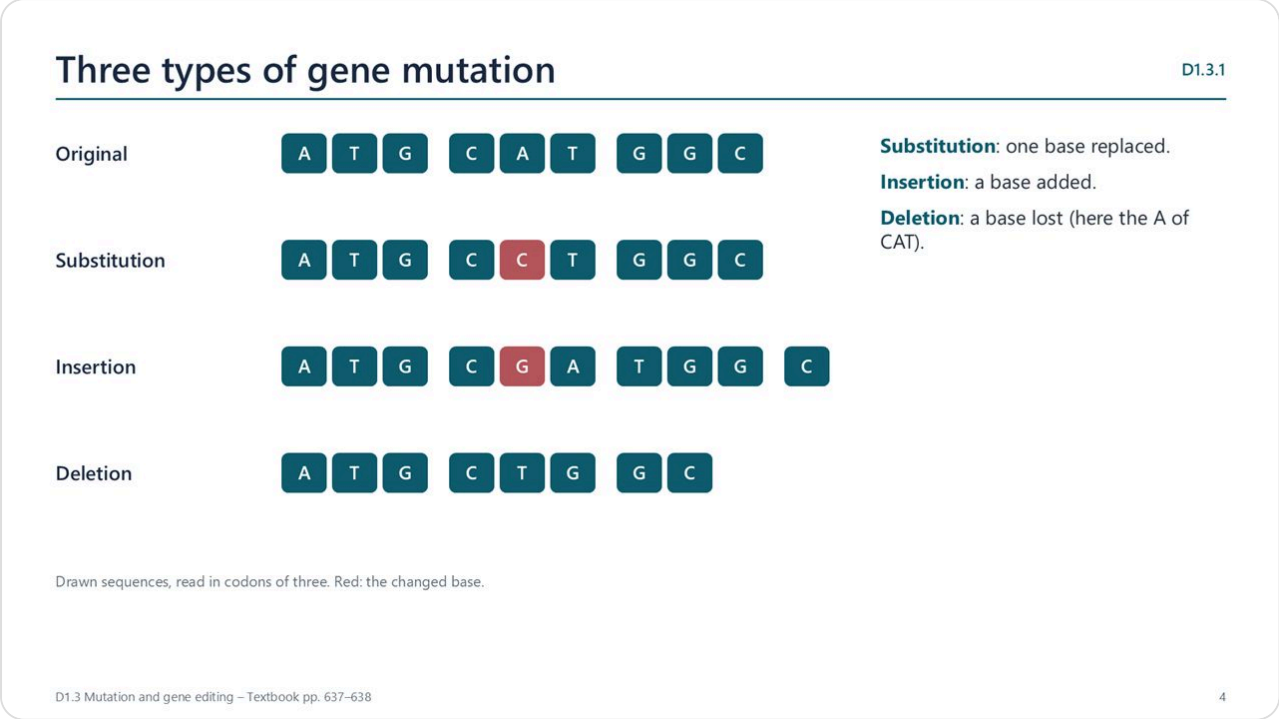
Notes written for this site. Slides, videos, worksheets and self-check questions: whatthecell.org/subtopics/sl-d1-3.html

Written for this site from the course textbook and class slides, with page references for each statement. The chapter is on pages 637–642 of the course textbook. Image credits are on the last slide of the slides for this subtopic.

What changes, and what follows

D1.3.1 Gene mutations are changes to a base sequence

pp. 637–638



In a **substitution**, one base is replaced by another. In an **insertion**, extra bases are added, and in a **deletion**, bases are lost.

D1.3.2 What base substitutions can lead to

p. 638

A **single-nucleotide polymorphism (SNP)** is a position where one base varies between individuals, and results from a substitution. Because the genetic code is degenerate, a substitution may not change the amino acid at all. If it does, usually only one amino acid changes. Rarely, a new stop codon ends the polypeptide early.

D1.3.3 What insertions and deletions can lead to

pp. 638–640

Bases are read in threes, so adding or losing one base shifts every codon that follows. This **frameshift** usually stops the polypeptide from working.

Gene	Change	Effect
HTT	Insertion of extra CAG repeats	40 or more repeats cause Huntington's disease, which damages neurons
CCR5	Delta 32: deletion of 32 bases	The receptor does not work, so people with two copies have strong resistance to most HIV strains

Watch out: the HTT repeat adds three bases at a time, so it keeps the reading frame. It adds extra amino acids rather than causing a frameshift.

Causes

D1.3.4 What causes gene mutations

pp. 640–641

Mutagens raise the rate of mutation. They include chemicals, such as benzene, asbestos and substances in tobacco smoke, and forms of radiation, such as short-wave UV, X-rays, gamma rays and alpha particles. Mutations also result from errors in DNA replication and repair.

D1.3.5 The random nature of mutation

p. 641

Mutations can occur anywhere in a genome, although some bases are more likely to mutate than others. No natural mechanism is known for changing a particular base in order to change a trait.

Watch out: organisms do not mutate because they need to. Antibiotics do not cause resistance mutations. They kill those bacteria without them.

Consequences

D1.3.6 Germ cells compared with somatic cells

p. 642

	Germ cells	Somatic cells
Inherited?	Yes, by offspring	No
Spread	May be in every cell of the offspring	Only in cells descended from the mutated cell
Main risk	Genetic disease	Cancer, when genes that control cell division are affected

D1.3.7 The source of all genetic variation

p. 642

A new **allele** arises only by mutation. Meiosis and fertilization mix existing alleles, but they do not create new ones. Most mutations are harmful or neutral for an individual, but in the long term they are essential for evolution by natural selection.

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